ECG Feature-Based Classification of Heart Disease Using Hybrid Ensemble Models

Sam Navin M[1]*, P.V.K. Ravi[2], C. Bhavitha[3] and S. Sanath[4]

[1,2,3,4] School of Computing, Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology, Avadi, Chennai, India
drsamnavinm@veltech.edu.in*

Abstract. Heart disease continues to be the primary cause of death globally. It highlights the importance of reliable and efficient diagnostic methods to reduce its impact. This paper presents a state-of-the-art ensemble-based machine learning architecture for predicting heart disease, aiming to enhance diagnostic accuracy. The proposed model learns ECG image-derived metadata, with a focus on ST elevation, T-wave inversion, and abnormal QRS complexes, as these are the most significant indicators of heart disease. To achieve exceptional prediction accuracy, the proposed work combines Random Forest (89%), XGBoost (89%), and CatBoost (90%) into a Soft Voting Classifier (91%), ensuring reliability by harnessing the distinct advantages of each model. The methodology includes various data manipulation techniques, label encoding, dataset splitting, and data cleaning to ensure strong performance across datasets. The soft voting classifier not only expedites diagnosis but also provides valuable insights, empowering patients to take proactive steps and enhance their heart health. The machine learning ensemble, featuring a Soft Voting Classifier with 91% accuracy, facilitates quicker and more precise heart disease diagnosis by promoting dependable, scalable medical decision-making.

Keywords: Heart Disease Prediction, Ensemble Machine Learning, Soft Voting Classifier, ECG Metadata Analysis, Random Forest, XGBoost, CatBoost, Diagnostic Accuracy, Medical Decision Support

1 Introduction

The World Health Organization (WHO) reports that cardiovascular diseases (CVDs) account for nearly 17.9 million deaths each year, ranking as the foremost cause of global mortality. This concerning statistic highlights the urgent need for diagnostic methods that are not only precise and effective but also scalable and easily deployable, particularly in efforts to improve clinical outcomes and reduce mortality rates **[1][6][12]**. Traditional diagnostic approaches such as echocardiography and angiography tend to be invasive, time-consuming, and dependent on advanced medical expertise, which can limit their utility in resource-constrained settings **[3][11].** In recent years, significant progress in artificial intelligence (AI) and machine learning (ML) has transformed medical diagnostics, introducing automated and cost-efficient systems that enhance accuracy and diagnostic speed **[2][6][15]**. ML techniques are particularly effective in recognizing complex patterns in large datasets, facilitating early and accurate detection of heart-related issues **[8][13]**. Ensemble

G. Dilip et al. (eds.), *Proceedings of the International Conference on Intelligent Systems and Digital Transformation (ICISD 2025)*, Atlantis Highlights in Intelligent Systems 15,
https://doi.org/10.2991/978-94-6463-866-0_78

learning models, such as Random Forest and XGBoost, have proven especially successful in modeling healthcare data, due to their ability to capture intricate feature interactions, especially within ECG-derived metadata **[1][12][16]**. The hybrid ensemble Soft Voting Classifier combines the predictive strengths of Random Forest, XGBoost, and CatBoost. The model targets key diagnostic indicators extracted from ECG metadata, including pathological Q waves, QRS complex abnormalities, T wave inversions, and ST segment elevations for detecting CVDs **[10][18]**. To boost the model's robustness and prevent overfitting, a comprehensive preprocessing pipeline was implemented, involving label encoding, normalization, and strategic data splitting **[3][9]**. To ensure ease of adoption in clinical practice, the system includes a Streamlit-based interface designed for real-time cardiac health analysis **[15][18]**.

Based on the analysis, CVDs are the top cause of death worldwide, emphasizing the need for effective and widely applicable diagnostic tools. Machine learning techniques, especially ensemble models like Random Forest, XGBoost, and CatBoost, significantly improve early detection using ECG data. The presence of a Streamlit interface ensures real-time usability in clinical environments, enhancing accessibility and effectiveness.

2 Literature Work

Ensemble learning techniques, including Random Forest, XGBoost, and AdaBoost, have been used for the early detection of heart diseases. Among these, a fine-tuned XGBoost model optimized using Bayesian methods and interpreted with SHAP presented the highest performance, indicating potential for real-world clinical deployment **[1]**. Machine learning significantly advances healthcare by enabling early identification of cardiovascular disorders and providing predictive, personalized care solutions by extending its utility to areas like epidemic prediction and operational improvement **[2]**. ML and SVM classifiers are used to detect cardiac irregularities and QRS complexes **[3]**. The potential of the ensemble learning model was examined through soft voting to boost the accuracy of heart disease prediction. By integrating Logistic Regression, SVM with RBF (Radial Basis Function Kernel, and Random Forest, the ensemble model delivers better diagnostic results than single models. The method strengthens the clinical diagnosis by achieving higher precision, recall, and AUC, supporting timely detection and improved patient outcomes **[4]**. The work applies machine learning techniques to detect CVD, using feature extraction methods like SMOTE (Synthetic Minority Oversampling Technique) and mean substitution. SVM and ensemble learning are evaluated, with hard voting achieving 93% accuracy. L1 regularization was employed to minimize overfitting and bias.**[5]**. In CVG, ML, and AI facilitated better diagnosis by overcoming data complexity and variability between observers, improving the reliability of image-based interpretations **[6]**. When applied to predict mortality after myocardial infarction, ML models delivered more precise risk assessments than logistic regression, particularly in high-risk groups **[7]**.

A combination of CNN, BiLSTM, and attention mechanisms achieved 94.28% accuracy in classifying cardiovascular and congenital conditions using IoMT-generated data, though the results suggested the need for validation across multiple datasets **[8]**. Comparisons between ST-elevation myocardial infarction

(STEMI) and non-ST-elevation myocardial infarction (NSTEMI) in elderly populations indicated lower cardiovascular mortality in non-revascularized NSTEMI cases, emphasizing the importance of further clinical evaluation **[9]**. A new ECG-derived metric, Q-wave area (QWA), showed a strong association with non-viable heart tissue, suggesting its usefulness in assessing myocardial viability **[10]**. Challenges in interpreting athletes' ECGs, especially T-wave abnormalities, highlighted the need for standardized diagnostic guidelines to minimize false positives **[11]**. Ensemble learning models benefited from feature selection and optimization processes, which enhanced their predictive accuracy and affirmed the applicability of ML in clinical diagnostics **[12]**. CNN architectures performed well in ECG-based classification tasks, accurately identifying complex patterns, though further clinical testing was recommended **[13]**. To manage imbalanced data and improve model outcomes, researchers applied hierarchical deep learning combined with GAN-generated synthetic samples **[14]**. The iCardo platform focused on the scalable deployment of ML-based cardiac diagnostics, emphasizing the importance of preprocessing and prediction efficiency **[15]**. Multiple classifier ensemble systems further confirmed the reliability of blended approaches in medical prediction tasks **[16]**. Transformer models adapted for visual analysis of ECGs exhibited strong capabilities in identifying nuanced cardiac patterns, indicating promising diagnostic applications **[17]**. Novel models like EnsCVDD-Net and BICVDD-Net achieved high accuracy, demonstrating the flexibility and effectiveness of ensemble learning in CVDs detection **[18]**.

From the above literature, it is obvious that ML has played a transformative role in enhancing the early detection of CVDs. Ensemble algorithms like Random Forest, XGBoost, and CatBoost have shown strong predictive performance, while complementary methods such as data balancing and feature extraction further improve accuracy and model reliability. When these models are integrated with real-time user interfaces and detailed data visualizations, they provide accurate predictions and offer clinical usability. The driving motivation behind this work is to develop machine learning-based diagnostic tools that are accurate, scalable, and interpretable, contributing to the global effort to combat CVDs. This study presents three major contributions. First, it introduces a reliable diagnostic system that uses the ECG image metadata and combines multiple machine learning models through a Soft Voting ensemble strategy, resulting in improved prediction of heart conditions. Second, it incorporates a wide range of visualization tools, including class distribution plots and correlation matrices, to improve understanding of the dataset and model performance. Third, a Gradio-based interface was developed for real-time ECG image classification and prediction, offering instant diagnostic feedback. It extracts key features from uploaded samples and supports integration with cloud and IoMT systems for scalable deployment. Together, these contributions aim to provide a practical, accurate, and interpretable solution for real-world clinical applications.

3 Data and Methods

This sector outlines the structure of the proposed architecture and provides a step-by-step explanation of data collection, preprocessing techniques, and the implementation of the soft voting classification method.

3.1 Proposed Architecture

The proposed architecture provides a step-by-step workflow that transforms ECG reports into meaningful clinical features for classification. The proposed flow was presented in **Fig. 1**. It integrates advanced ensemble algorithms to improve predictive accuracy across different cardiac categories. The design ensures both computational efficiency and medical relevance through careful feature engineering and model evaluation.

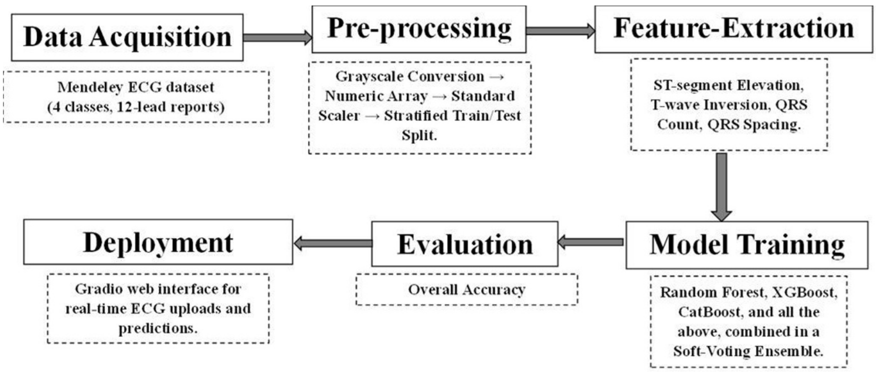

Fig. 1. Workflow of the proposed ECG-based heart disease classification system. The process begins with data acquisition from the Mendeley ECG dataset, consisting of 12-lead ECG reports categorized into four diagnostic classes. Preprocessing involves grayscale conversion, normalization, and stratified data splitting. Clinically relevant features such as ST-segment elevation, T-wave inversion, QRS count, and spacing are extracted. These features are used to train an ensemble of machine learning models, Random Forest, XGBoost, and CatBoost, combined through soft voting. The system's performance is evaluated using classification metrics and feature-importance visualizations, ensuring accurate and interpretable outputs.

3.2 Data Acquisition

The ECG dataset employed in this work was obtained from Mendeley Data, created by Ali Haider Khan and Muzammil Hussain at the Ch. Pervaiz Elahi Institute of Cardiology in Multan, Pakistan **[19]**. It contains a total of 928 ECG reports, categorized into four groups like individuals with normal heart function, patients exhibiting abnormal heart rhythms, those with a prior history of myocardial infarction (MI), and current MI cases. Each report corresponds to a 12-lead ECG layout, a diagnostic standard in cardiology that records the heart's electrical activity from twelve perspectives, mainly, leads I, II, III, aVR, aVL, aVF, and V1 through V6. These leads capture electrical impulses from different angles, helping detect important indicators such as ST-segment elevation, T-wave inversion, and QRS abnormalities. These features are instrumental in training an ensemble-based machine learning model aimed at achieving precise and early detection of various cardiac conditions. The dataset plays a key role in building diagnostic systems that are accurate and scalable. **Table 1** presents the distribution of samples across each class along with the corresponding number of lead-level segments. **Fig. 2** displays the sample ECG report of a normal person, highlighting the dataset's breadth and diagnostic relevance.

Table 1. Table captions should be placed above the tables.

Category	No. of ECG Reports	Leads per Report	Total Lead Level Segments
Normal Person ECG Images	284	12	3408
Abnormal Heartbeat Patients	233	12	2796
Myocardial Infarction Patients	240	12	2880
History of MI (Recovered Patients)	172	12	2064

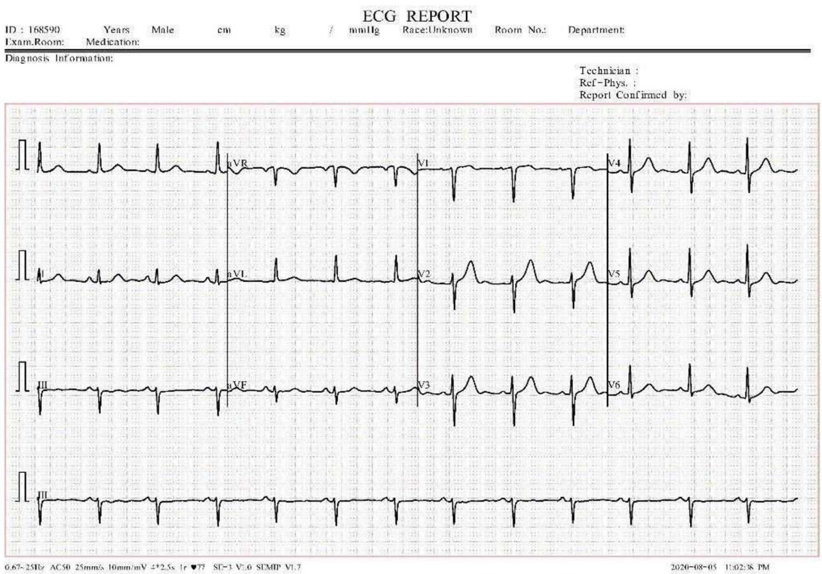

Fig. 2. Sample ECG report of a normal person, showing consistent waveform patterns across all 12 leads. This category includes 284 ECG samples, totaling 3408 individual lead traces.

3.3 Pre-processing and Feature Extraction

ECG samples obtained from the Mendeley dataset experience a structured preprocessing workflow before model training. Each sample was first transformed into grayscale and then converted into a numerical array by enabling efficient data handling. From these arrays, ST-segment elevation, T-wave inversion, QRS complex frequency, and spacing are extracted and compiled into a feature matrix. To ensure uniformity across inputs and optimize model performance, the features are normalized using z-score standardization through the StandardScaler. The processed data was divided into training and testing sets in an 80:20 ratio, with stratified sampling applied to preserve the proportional distribution of each diagnostic class. This ensures balanced representation during learning and consistent generalization across all categories.

3.4 Model Training

The training strategy in this work involves building a soft voting ensemble classifier that combines the strengths of three diverse machine learning models, Random Forest, XGBoost, and CatBoost, to improve the accuracy and robustness of heart disease prediction based on features extracted from ECG samples.

- **Random Forest.** Random Forest was used due to its effectiveness in handling nonlinear relationships and high-dimensional feature spaces. By building several decision trees and combining their predictions, it functions as an ensemble learning technique that increases accuracy. Our work advanced significantly from this approach since it minimizes overfitting and works well with unbalanced datasets. It enhances generalization, making it particularly suitable for complex ECG data and imbalanced class distributions.

- **XGBoost.** Extreme Gradient Boosting, or XGBoost, was incorporated into our model due to its effectiveness, speed, and high predicted accuracy. It uses boosting approaches to help weak learners one step at a time and performs significant work on structured medical data. The integrated regularization algorithms in XGBoost decreased overfitting while maintaining excellent precision for ECG classification. It was possible to train faster and achieve higher accuracy with a large dataset by processing it efficiently.
- **CatBoost.** CatBoost was used to increase the accuracy when dealing with categorical values, which reduced the amount of preprocessing. It handles categorical variables and was especially used for our ECG dataset. It increased the stability and predictive power of the model, especially in detecting subtle variations in ECG that corresponded with different heart diseases. With gradient-based decision trees and the reduced training time,

CatBoost strengthened the ensemble classifier, and predictions remained both fast and reliable across different ECG patterns.

- **Soft Voting.** Soft voting was crucial in this work because it combines the predictive probabilities of individual classifiers, such as Random Forest, XGBoost, and CatBoost, to make a final decision. This is different from hard voting, which relies on majority voting. In this work, the soft voting considers the confidence level of each model for different classes, thus making the predictions more robust and accurate. This helps aggregate the probabilities such that the Voting Classifier capitalizes on the strengths of each algorithm. Soft Voting improves the overall performance of the cardiac health prediction system.

This work trained a soft voting ensemble model by integrating Random Forest, XGBoost, and CatBoost to improve heart disease prediction from ECG-derived features. Each algorithm contributed distinct strengths in handling data complexity, regularization, and feature interactions. The soft voting mechanism aggregated probabilistic outputs, resulting in a more accurate and reliable cardiac risk classification system.

3.5 Model Deployment using Gradio Interface

In the final phase of this work, a user-friendly interface was implemented using Gradio to facilitate real-time ECG sample prediction through the previously trained ensemble classification model. Users can upload a standard 12-lead ECG image, which automatically converted to grayscale and numerically processed to extract critical features such as ST-segment elevation, T-wave inversion, QRS complex count, and interval spacing. These features are standardized using a pre-trained scaler to align with the model's input specifications. The normalized data was then passed to a Soft Voting Ensemble classifier that integrates the outputs of Random Forest, XGBoost, and CatBoost models to produce a final prediction. This prediction was mapped to corresponding clinical categories, such as Normal or Myocardial Infarction, and presented to the user instantly. This Gradio-based interface was designed for accessibility and ease of use, making it a valuable diagnostic support tool for clinicians and researchers. It offers potential for seamless integration with cloud infrastructure and IoMT ecosystems, enabling scalable, remote cardiac monitoring in real-world healthcare environments.

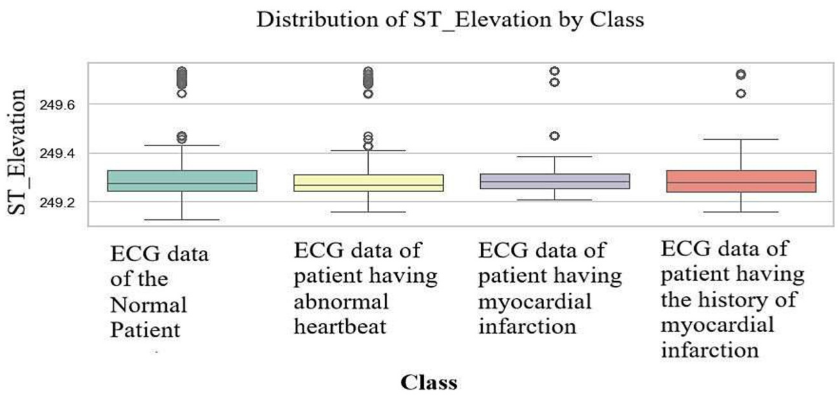

Fig. 3. Distribution of ST_Elevation by Class. These box plots show feature distributions. ST_Elevation medians range ~249.4-249.5 m, with outliers indicating variability.

4 Results and Analysis

We successfully executed the implementation in Jupyter Lab (V. 4.2.5) using Python (V. 3.12.3), and the results were generated based on ECG image-derived metadata. The distributions of four key features, ST_Elevation, T_Inversion, QRS_Count, and QRS_Spacing, were visualized in **Fig. 3, 4, 5, and 6**, providing insight into their diagnostic relevance.

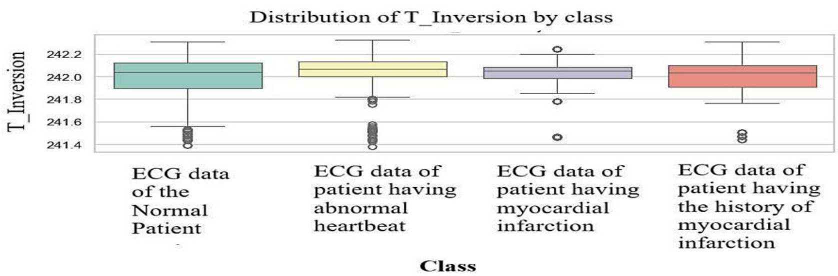

Fig. 4. Distribution of T_Inversion by Class. These box plots show feature distributions where T_Inversion varies by class, ranges 241.8 -242.2 units, with outliers indicating variability.

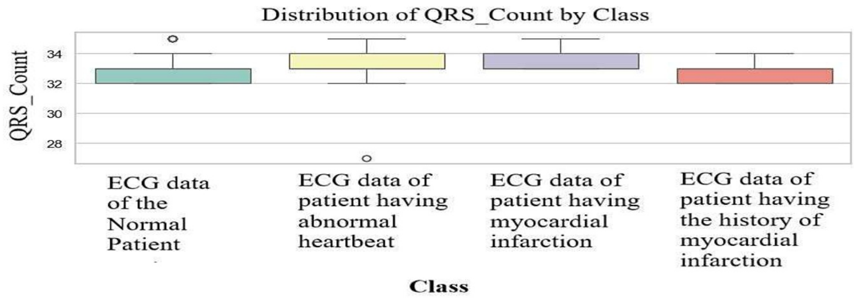

Fig. 5. Distribution of QRS_Count by Class. These box plots show feature distributions, where QRS_Count averages 2-5 peaks, with outliers indicating variability.

ST_Elevation (STE) represents the amplitude of the ST segment in an ECG signal. An elevated ST segment is a key indicator of acute myocardial infarction, making this metric vital for identifying severe cardiac events. The T-Inversion (TI**)** signifies the inversion of the T-wave, which indicates myocardial ischemia, ventricular stress, or abnormal repolarization. Changes in this feature assist in differentiating various cardiac conditions. The QRS_Count (QRSC) specifies the total number of QRS complexes (heartbeats) within a defined time window by correlating with heart rate. Deviations from the normal count can suggest arrhythmias or irregular cardiac rhythms. The QRS_Spacing (QRSS) measures the interval between consecutive QRS complexes, offering insight into heartbeat consistency. Variability in this spacing may reveal potential rhythm disorders or conduction abnormalities. Together, these features play a vital role in identifying and classifying heart conditions, forming the core input for the machine learning models used in this study. **Fig. 7** presents the correlation matrix of ECG-derived features across all classes. It reveals a moderate positive correlation between ST_Elevation and T_Inversion. These relationships help in identifying relevant features and reducing redundancy, thereby enhancing model performance and interpretability.

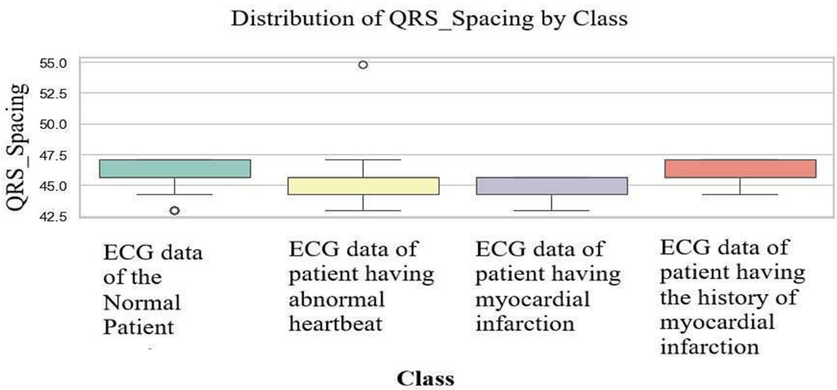

Fig. 6. Distribution of QRS_Spacing by Class. These box plots show feature distributions where QRS_Spacing ranges ~30-50 units, with outliers indicating variability.

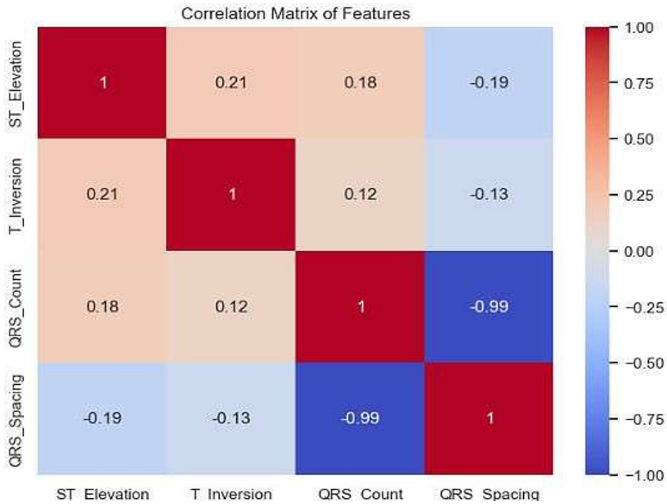

Fig. 7. Correlation matrix of ECG-derived features across all classes. This heatmap shows correlations. ST_Elevation and T_Inversion may have a moderate positive correlation (~0.3-0.5), while QRS_Count and QRS_Spacing might be negatively correlated (~-0.2), assisting the feature selection.

Table 2. Accuracy Assessment.

Algorithms	Accuracy (%)	Misclassification Rate
Random Forest	89	11
XGBoost	89	11
CatBoost	90	10
Soft Voting Classifier	91	9

Table 2 summarizes the performance of four classification algorithms used in the study. Both Random Forest and XGBoost achieved 89% accuracy with an 11% error rate, while CatBoost performed slightly better with 90% accuracy. The Soft Voting Classifier outperformed all individual models, achieving 91% accuracy and the lowest error rate of 9%. These results highlight the strength of ensemble learning in enhancing the accuracy of cardiac condition prediction. **Fig.8** displays an ECG Heart Disease Predictor tool evaluating two ECGs using a Soft Voting Ensemble classifier. A Gradio-based interface facilitates real-time classification, detecting one as normal and the other as myocardial infarction. This tool processes images instantly, offering precise diagnostic insights for cardiovascular health assessment.

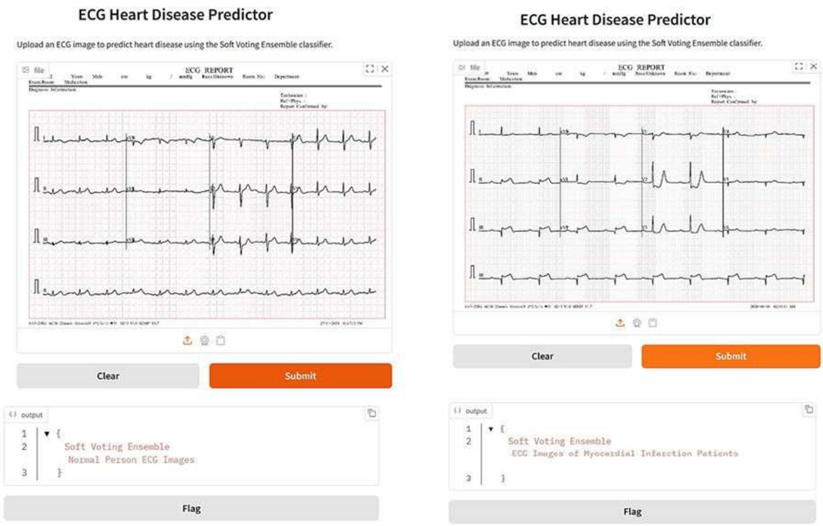

Fig. 8. ECG Heart Disease Predictor tool. It evaluates two ECG samples using a Soft Voting Ensemble classifier and predicts accordingly.

5 Discussion

This research contributes to the classification of cardiac disorders by proposing a hybrid ensemble-based machine learning approach that uses ECG-derived metadata. The soft classifier model integrates Random Forest, XGBoost, and CatBoost, demonstrating high accuracy in detecting key indicators of heart disease. A real-time diagnostic interface is also incorporated, broadening the accessibility and practicality of AI-powered healthcare tools. Despite its effectiveness, the ensemble strategy carries a risk of overfitting, particularly with smaller datasets, a limitation partially addressed through soft voting. Future investigations should focus on validating the model's performance across diverse demographic groups, expanding the dataset, and incorporating explainable AI methodologies to enhance clinical transparency and trust. Potential enhancements include the use of advanced deep learning architectures such as Convolutional Neural Networks (CNNs) for automated feature extraction and Generative Adversarial Networks (GANs) for synthetic ECG data generation. Deployment on cloud-based platforms can facilitate scalable, real-time applications, while integration with Internet of Medical Things (IoMT) devices can enable immediate, point-of-care diagnostics, boosting the system's effectiveness and accessibility in varied healthcare contexts.

6 Conclusion

This work presents a cardiac health prediction system that leverages machine learning to effectively assess heart conditions. By combining the strengths of Random Forest, XGBoost, CatBoost, and a Soft Voting Classifier, the ensemble model delivers robust performance. The Soft Voting Classifier achieved the highest precision (91%) by demonstrating the advantage of integrating multiple algorithms. To enhance usability, a Gradio-powered interface was developed in the final phase, enabling real-time ECG image classification with instant diagnostic output. The interface extracts key features from uploaded ECG images and provides accurate predictions, supporting seamless integration with cloud platforms or IoMT-based healthcare systems. Future enhancements may include incorporating additional features, deploying deep learning models, and scaling the solution for real-time clinical use in AI-driven healthcare.

References

1. Mienye, I. D., & Jere, N. (2024). Optimized ensemble learning approach with explainable AI for improved heart disease prediction. Information, 15(7), 394.
2. Banapuram, C et al. (2024). A Comprehensive Survey of Machine Learning in Healthcare: Predicting Heart and Liver Disease, Tuberculosis Detection in Chest X-Ray Images. SSRG International Journal of Electronics and Communication Engineering, 11(5), 155-169.
3. Rizwan, A. et al. (2022). A machine learning approach for the detection of QRS complexes in electrocardiogram (ECG) using discrete wavelet transform (DWT) algorithm. Computational Intelligence and Neuroscience, 2022(1), 9023478.
4. Kim, B. J. (2024). Enhancing Heart Disease Prediction Accuracy through Soft Voting Ensemble Techniques. International Journal of Internet, Broadcasting and Communication, 16(3), 290-297.
5. Rajput, B., Nikhita, D., Veena, E., Mallibhat, K., Chikkamath, S., & Nirmala, S. R. (2024, March). Cardiovascular Disease Detection Using Ensemble Learning. In International Conference on Recent Trends in Machine Learning, IOT, Smart Cities & Applications (pp. 1-10). Singapore: Springer Nature Singapore.
6. Reza-Soltani, S. et al. (2024). The role of artificial intelligence and machine learning in cardiovascular imaging and diagnosis. Cureus, 16(9).
7. Khera, R. et. Al. (2021). Use of machine learning models to predict death after acute myocardial infarction. *JAMA cardiology*, *6*(6), 633-641.
8. Pachiyannan, P. et al(2024). A novel machine learning-based prediction method for early detection and diagnosis of congenital heart disease using ECG signal processing. *Technologies*, *12*(1), 4.
9. Chang, S. S. et al. (2022). Prognosis between ST-elevation and non-ST-elevation myocardial infarction in older adult patients. Frontiers in Cardiovascular Medicine, 8, 749072.
10. Kumpamool, P. et al. (2022). Prediction of nonviable myocardium by ECG Q-Wave parameters: A 3.0 T cardiovascular magnetic resonance study. Indian Heart Journal, 74(2), 105-109.
11. Torabi, A. J. et al. (2022). Athlete ECG T-wave abnormality interpretation patterns by non-experts. American Heart Journal Plus: Cardiology Research and Practice, 17, 100153.
12. Yarasuri, V. K., Dhumsapuram, S. R., Puligundla, S. M., Ramabhotla, V. S. K., Thupalli, N. V., & Nisha, K. L. (2022). Developing Machine Learning Models for Cardiovascular

Disease Prediction. In 2022 2nd Asian Conference on Innovation in Technology (ASIANCON) (pp. 1–6). IEEE.Amrita Vishwa Vidyapeetham

13. Khera, R. et. Al. (2021). Use of machine learning models to predict death after acute myocardial infarction. *JAMA cardiology*, *6*(6), 633-641.
14. Pachiyannan, P. et al(2024). A novel machine learning-based prediction method for early detection and diagnosis of congenital heart disease using ECG signal processing. *Technologies*, *12*(1), 4.
15. Chang, S. S. et al. (2022). Prognosis between ST-elevation and non-ST-elevation myocardial infarction in older adult patients. Frontiers in Cardiovascular Medicine, 8, 749072.
16. Kumpamool, P. et al. (2022). Prediction of nonviable myocardium by ECG Q-Wave parameters: A 3.0 T cardiovascular magnetic resonance study. Indian Heart Journal, 74(2), 105-109.
17. Torabi, A. J. et al. (2022). Athlete ECG T-wave abnormality interpretation patterns by non-experts. American Heart Journal Plus: Cardiology Research and Practice, 17, 100153.
18. Yarasuri, V. K., Dhumsapuram, S. R., Puligundla, S. M., Ramabhotla, V. S. K., Thupalli, N. V., & Nisha, K. L. (2022). Developing Machine Learning Models for Cardiovascular Disease Prediction. In 2022 2nd Asian Conference on Innovation in Technology (ASIANCON) (pp. 1–6). IEEE.Amrita Vishwa Vidyapeetham
19. Khan, Ali Haider; Hussain, Muzammil (2021), "ECG Images dataset of Cardiac Patients ", Mendeley Data, V2 (https://data.mendeley.com/datasets/gwbz3fsgp8/2).